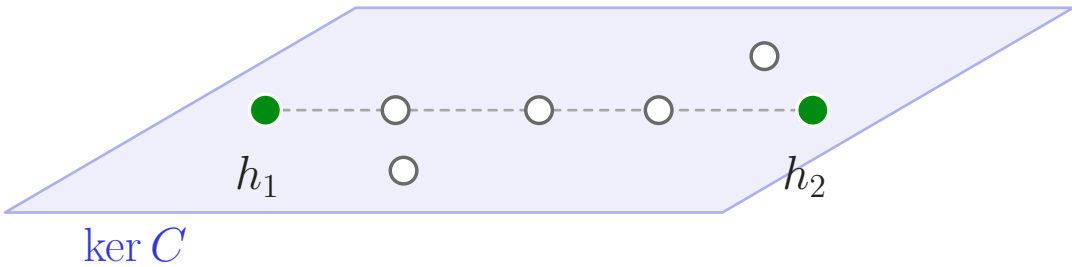


The family can separate compatible features

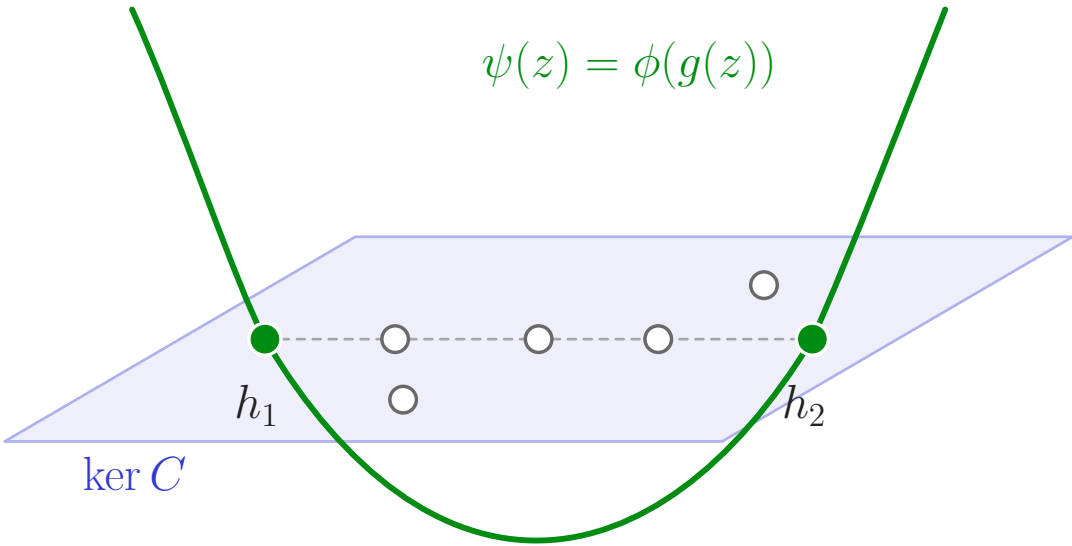
Feature-space schematic

Certificate alone



compatible features need not identify images

Certificate + family



separation is a property to establish

$$\underbrace{g(z) = \mu + Uz}_{\text{PCA: affine in image space}}$$

$$\underbrace{\psi(z) = \phi(\mu + Uz)}_{\text{can be curved in feature space}}$$

$$k < r - q, \qquad \{z : \psi(z) \in \text{col } H\} = \{z_1, \dots, z_N\}$$

fixed smooth family + span separation + exact-certificate conditions: generic identification